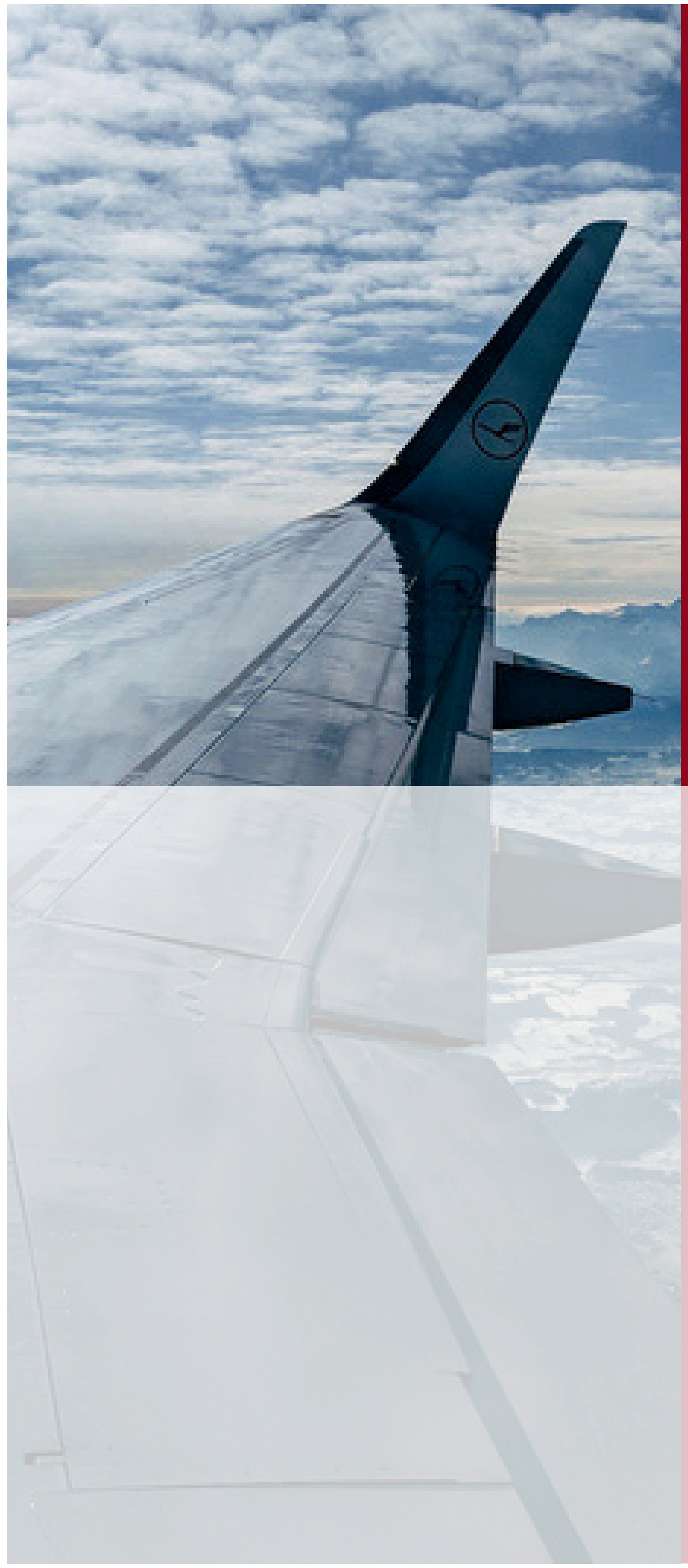
How Airport Delays Cascade Across the US

Sawyer McKenney
CSCI 4022

Background

- One delay can trigger a ripple effect
- Flight delays cost the economy $33B annually
- 78% of delays are system-driven, not external
- Standard analysis misses the network-level view



Research Questions

Which airports and routes are most influential to delays?

How do delays propagate through the network?

What combination of conditions reliably predicts large-scale cascades?

Data Source/Collection

- ~500K domestic flights per month
- Automated pipeline to extract data
- 12M+ flight records processed
- Data cleaned and validated
- Combined into a single parquet dataset for analysis

DEPARTMENT OF TRANSPORTATION
UNITED STATES OF AMERICA

```
port requests
port os

f download_bts_months(year_month_pairs, output_dir="bts_data"):
  os.makedirs(output_dir, exist_ok=True)

  base_url = ("https://transtats.bts.gov/PREZIP/" "On_Time_Reporting_Carrier_On_Time_Performance_1987_present_{}_{}.

  for year, month in year_month_pairs:
      url = base_url.format(year, month)
      filename = f"ontime_{year}_{month:02d}.zip"
      filepath = os.path.join(output_dir, filename)
      print(f"Downloading {year}-{month}")
      response = requests.get(url, stream=True)
      if response.status_code == 200:
          with open(filepath, "wb") as f:
              for chunk in response.iter_content(chunk_size=8192):
                  f.write(chunk)
          print(f"Saved {filename}")
      else:
          print(f"  FAILED {year}-{month} (status {response.status_code})")
April 2024 through December 2025 (12 months)
nths = ([(2024, m) for m in range(4, 13)] + [(2025, m) for m in range(1, 13)])

wnload_bts_months(months)
8m 4.9s
```

Data Cleaning

- Data is self-reported, which caused noise
- Missing Tail Number → filled with unknown
- Cancelled Flights → split into a separate df
- Diverted Flights → split into a separate df
- Remaining null → rows dropped
- Route level aggregation → grouped my origin
- Negative delays → kept (early departures)

```
FlightDate                              0
Reporting_Airline                       0
Tail_Number                         26083
Flight_Number_Reporting_Airline         1
Origin                                  0
Dest                                    0
CRSDepTime                              0
DepDelay                           163882
DepDelayMinutes                    163882
DepDel15                           163882
ArrDelay                           203742
ArrDelayMinutes                    203742
ArrDel15                           203742
Cancelled                               0
CancellationCode                 12251741
Diverted                                0
Distance                                0
CarrierDelay                      9768297
WeatherDelay                      9768297
NASDelay                          9768297
SecurityDelay                     9768297
LateAircraftDelay                 9768297
dtype: int64
```

```
df["Tail_Number"] = df["Tail_Number"].cat.add_categorie
df["CarrierDelay"] = df["CarrierDelay"].fillna(0)
df["WeatherDelay"] = df["WeatherDelay"].fillna(0)
df["NASDelay"] = df["NASDelay"].fillna(0)
df["SecurityDelay"] = df["SecurityDelay"].fillna(0)
df["LateAircraftDelay"] = df["LateAircraftDelay"].filln
df["CancellationCode"] = df["CancellationCode"].cat.add
df = df.dropna(subset=["Flight_Number_Reporting_Airline

print(df.isna().sum())
```

✓ 2.8s

```
FlightDate                         0
Reporting_Airline                  0
Tail_Number                        0
Flight_Number_Reporting_Airline    0
Origin                             0
Dest                               0
CRSDepTime                         0
DepDelay                      163882
DepDelayMinutes               163882
DepDel15                      163882
ArrDelay                      203742
ArrDelayMinutes               203742
ArrDel15                      203742
Cancelled                          0
CancellationCode                   0
Diverted                           0
Distance                           0
CarrierDelay                       0
WeatherDelay                       0
NASDelay                           0
SecurityDelay                      0
LateAircraftDelay                  0
dtype: int64
```

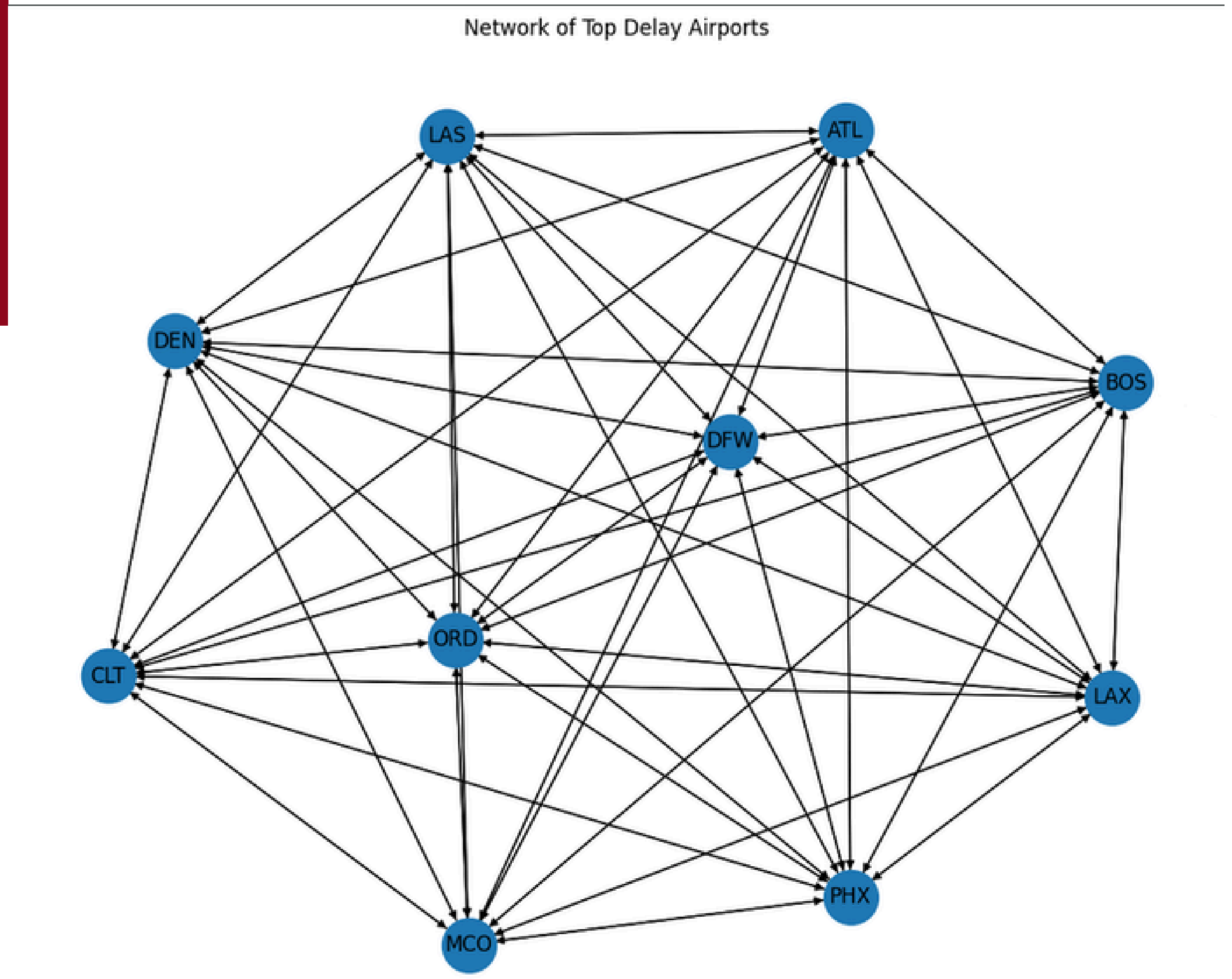
Graph

```python
# This is a directed graph
# nx is a library for working with graphs in python
Graph = nx.DiGraph()

for _, row in route_df.iterrows():
    Graph.add_edge(row["Origin"], row["Dest"], num_flights=row["num_flights"], avg_delay=row["avg_dep_delay"], pct_delayed=row["pct_delayed"])

print("Nodes:", Graph.number_of_nodes())
print("Edges:", Graph.number_of_edges())
```

✓ 0.0s

```
Nodes: 356
Edges: 7257
```

A graph was used to model the US airport system as a network, where delays flow between airports through flight routes

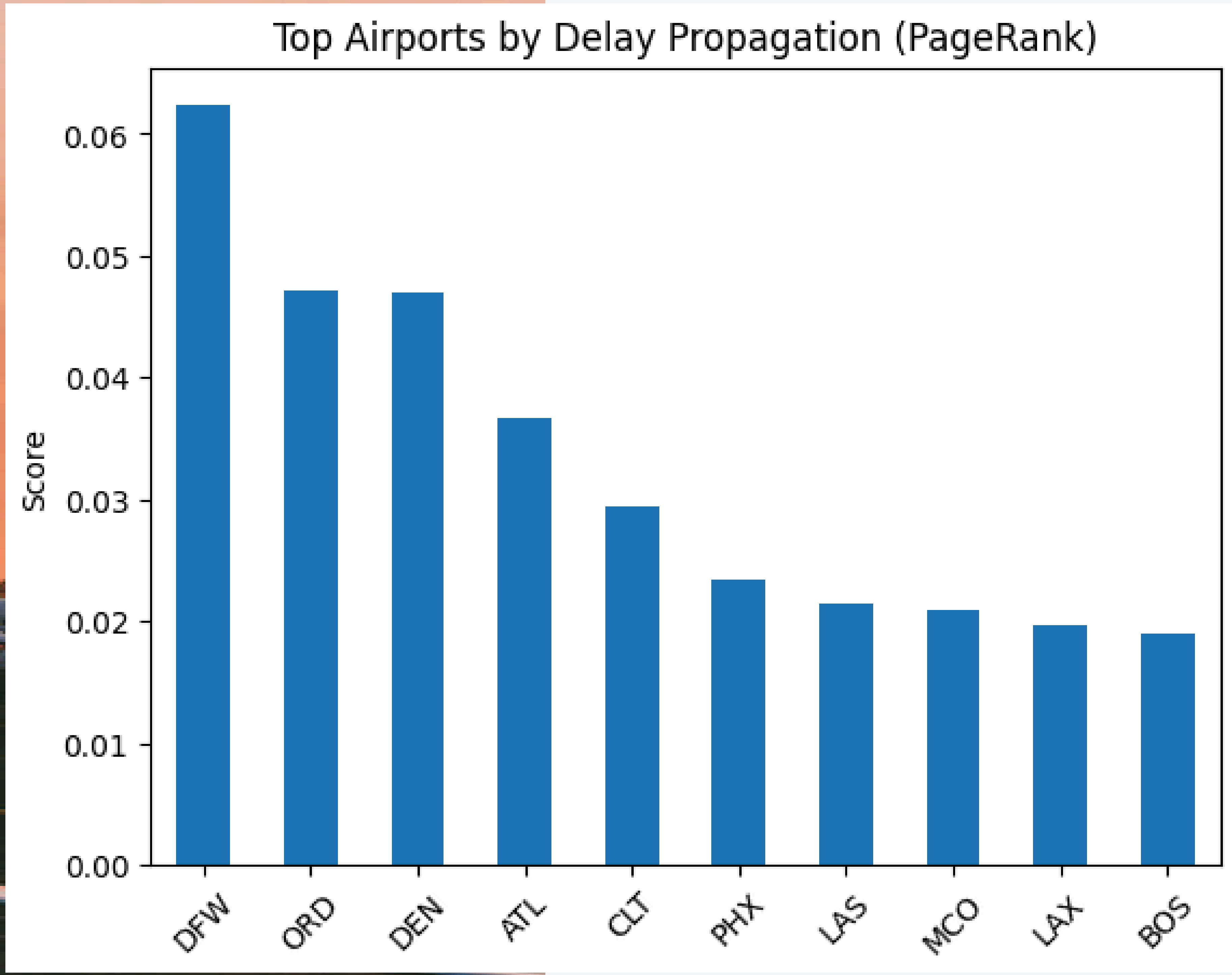


PageRank - Delay Propagation

Used to PageRank show how many delays at one airport spread throughout the network.

Each connection between airports was weighted by averaging the departure delay for that route.

Then, applied PageRank to score each airport based on how much delay it sends out through the network.



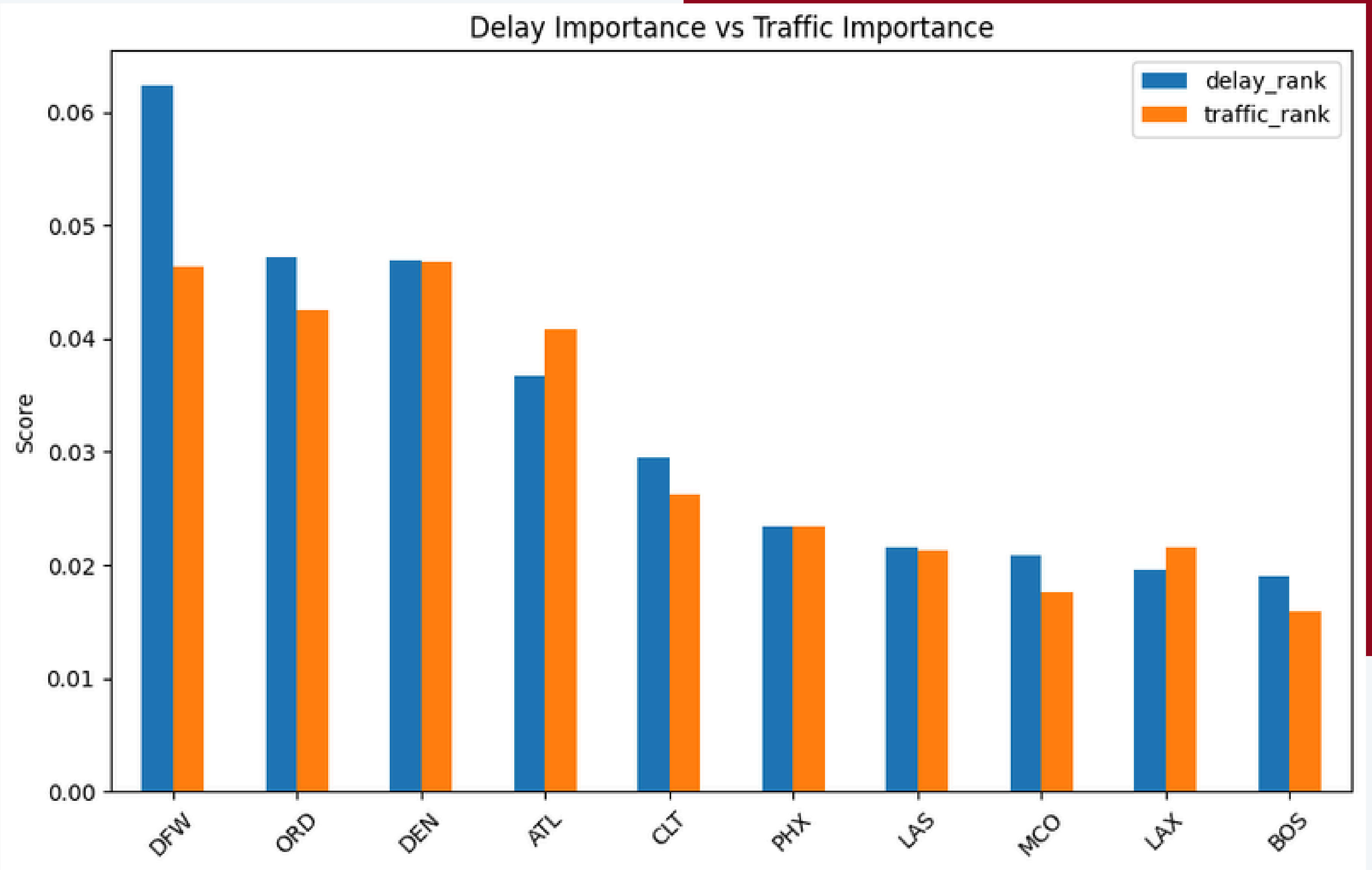
PageRank - Delay Propagation & Traffic Propigation

Traffic rank:

Tells us that airports are the backbone of the network, meaning that if these airports were not in the graph, they would create a gap in the network because they handle so much traffic.

Notice:

DFW has a large gap between traffic rank and delay rank. So its delays are disproportionate to its role in traffic.



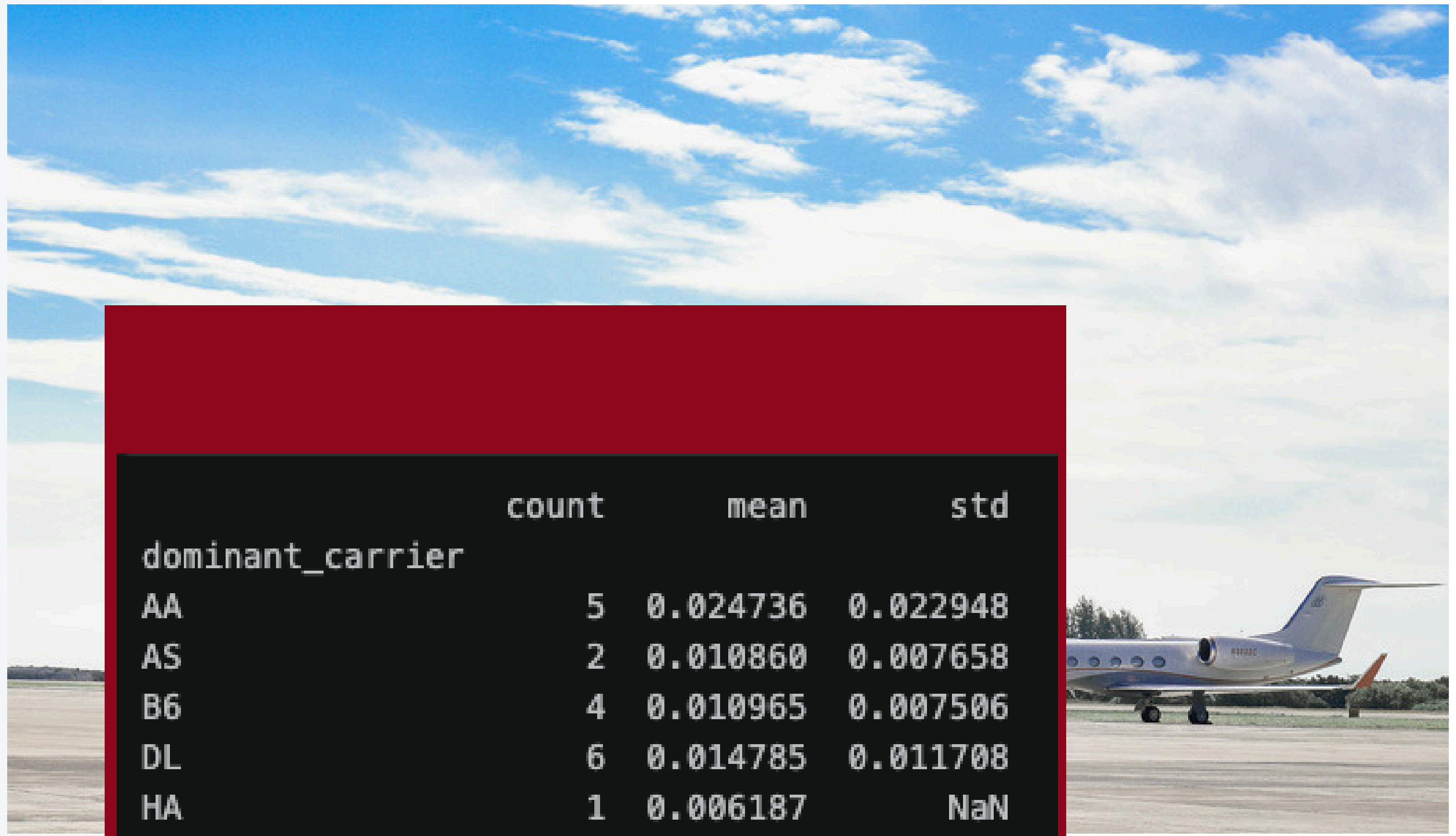
Do certain carriers' hubs have higher delays?

Question

Does it matter which airline runs the airport? If American runs DFW, do all American Hubs have high delay scores?

Findings

I counted how many hubs each carrier dominated (50K + flights), then found the average delay PageRank across those hubs, and then used standard deviation to determine how much the Delay PageRank varies across the hubs.



dominant_carrier	count	mean	std
AA	5	0.024736	0.022948
AS	2	0.010860	0.007658
B6	4	0.010965	0.007506
DL	6	0.014785	0.011708
HA	1	0.006187	NaN
NK	1	0.011944	NaN
UA	7	0.022159	0.017963
WN	21	0.008740	0.005983

Conclusion

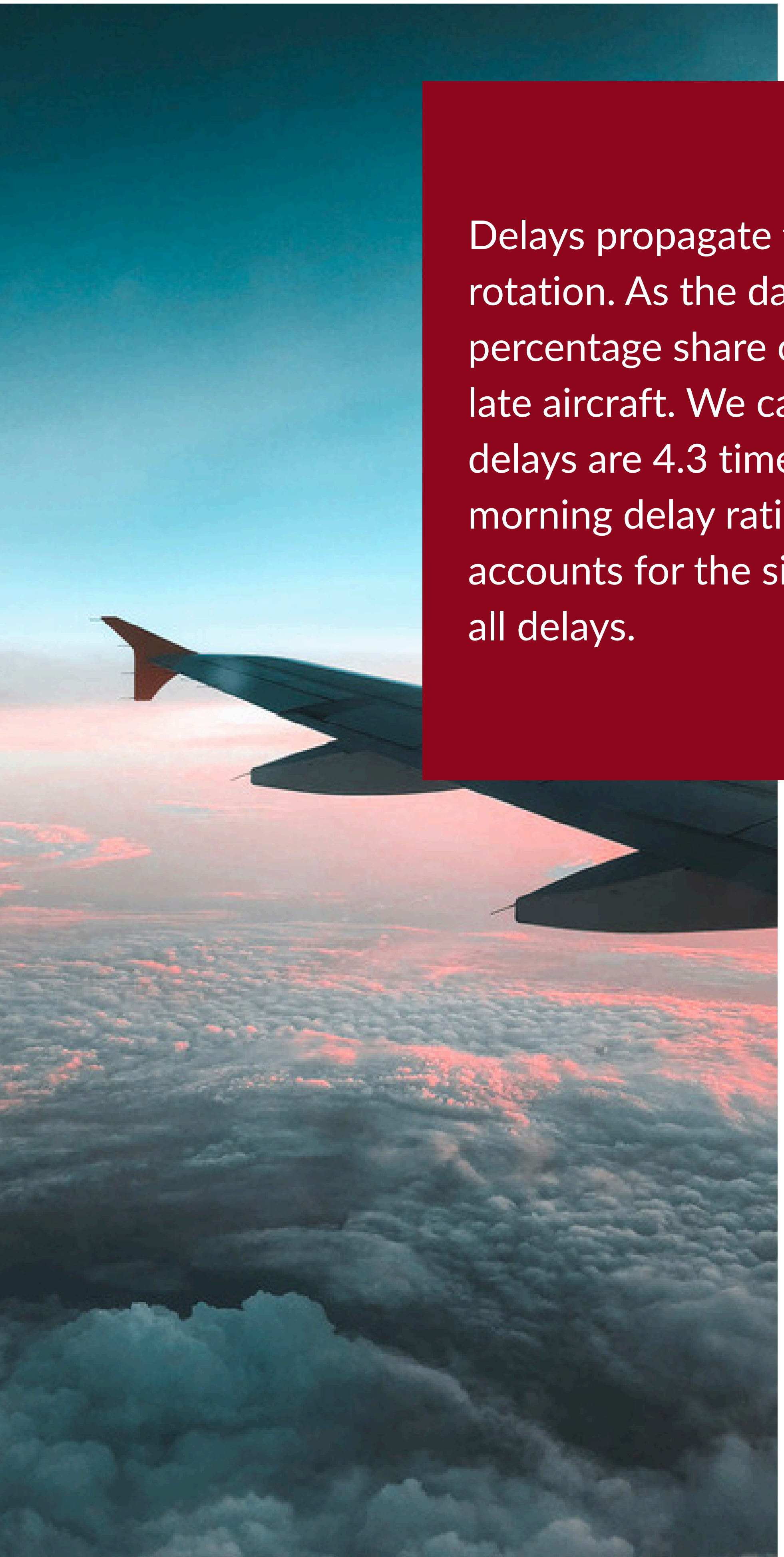
Carrier identity matters very little.

Which airports and routes are most influential to delays?

Influential airports: DFW, ORD, and DEN.

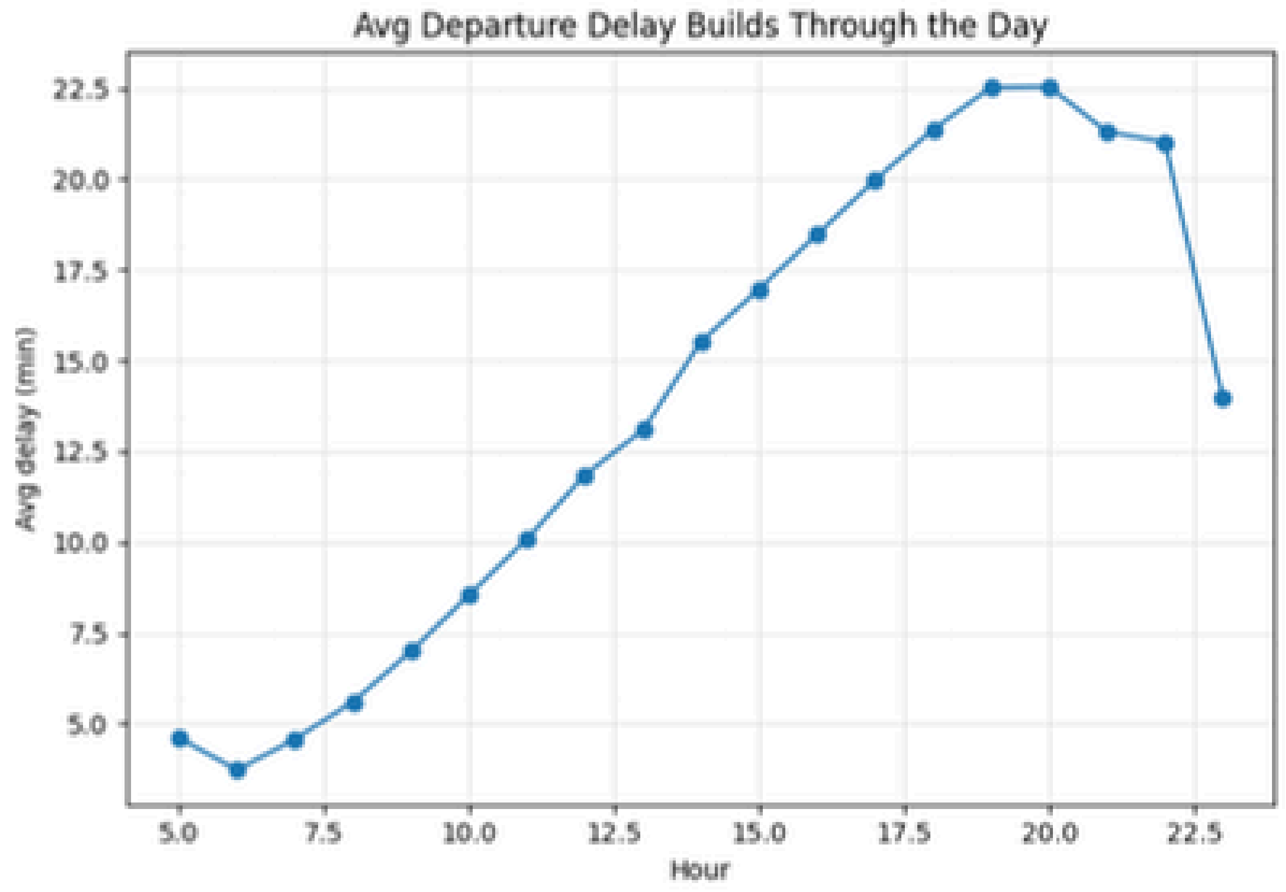
Influential routes (Three categories):

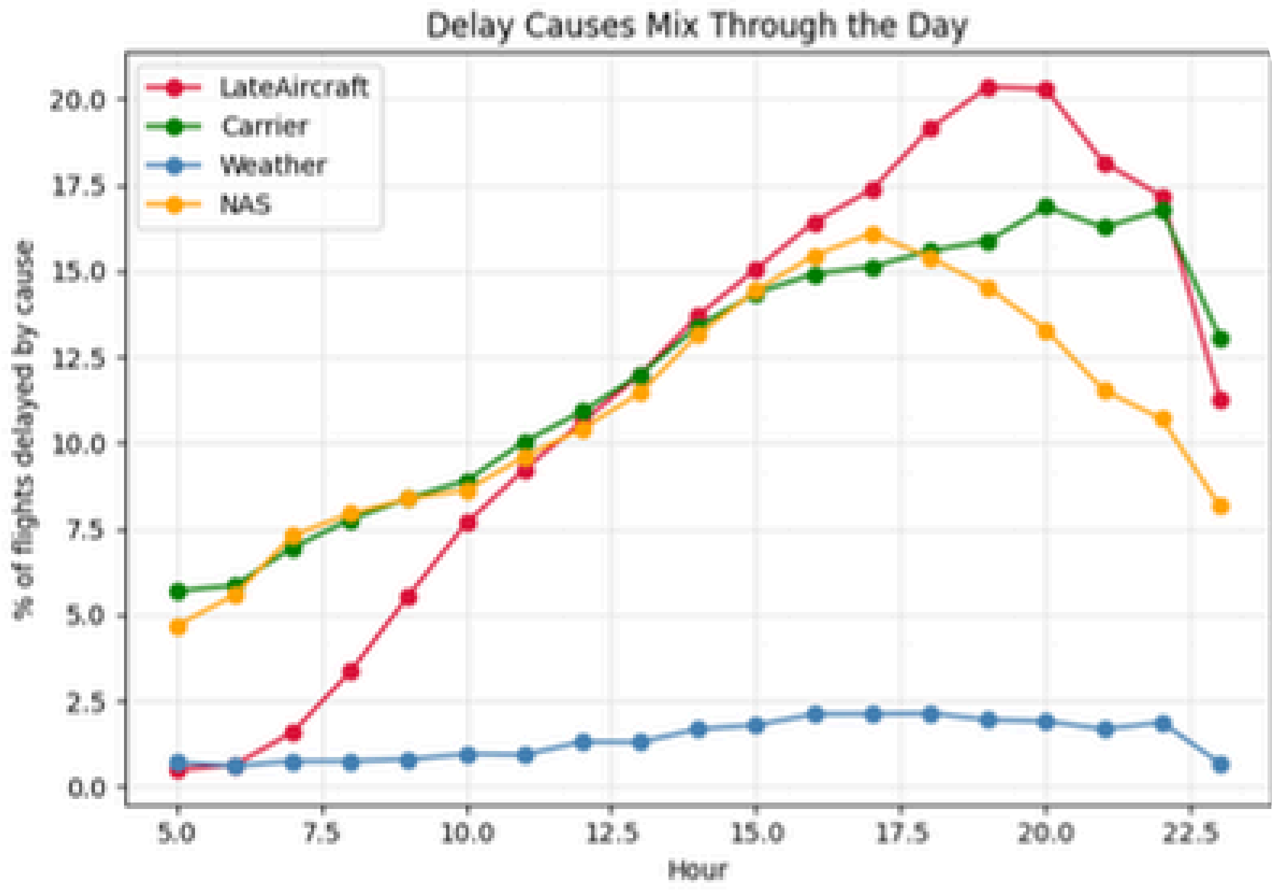
- High-volume corridors between hubs
- Niche routes that are constantly delayed (often regional airports to hubs)
- Carrier-specific underperformers (one airline that drags down a route)



How do delays propagate through the network?

Delays propagate through aircraft rotation. As the day progresses, the percentage share of flights delayed by late aircraft. We can see that evening delays are 4.3 times higher than the morning delay ratio. Late aircraft accounts for the single largest cause of all delays.





How cascade conditions are identified?

Convert each day into signals

- High weather delay
- High late aircraft delay
- High NAS, carrier delays
- Weekends/Peak patterns

Label days as cascades vs non-cascades via the 90th percentile of delay across all days

Apply Apriori and association rules

```
day_features = df_flights.groupby("FlightDate", observed=True).agg(
    avg_delay=("DepDelay", "mean"),
    total_delay=("DepDelay", "sum"),
    pct_weather_delay=("weather_flag", "mean"),
    pct_late_aircraft=("late_aircraft_flag", "mean"),
    pct_carrier_delay=("carrier_flag", "mean"),
    pct_nas_delay=("nas_flag", "mean"),
    num_flights=("DepDelay", "size"),
)
```

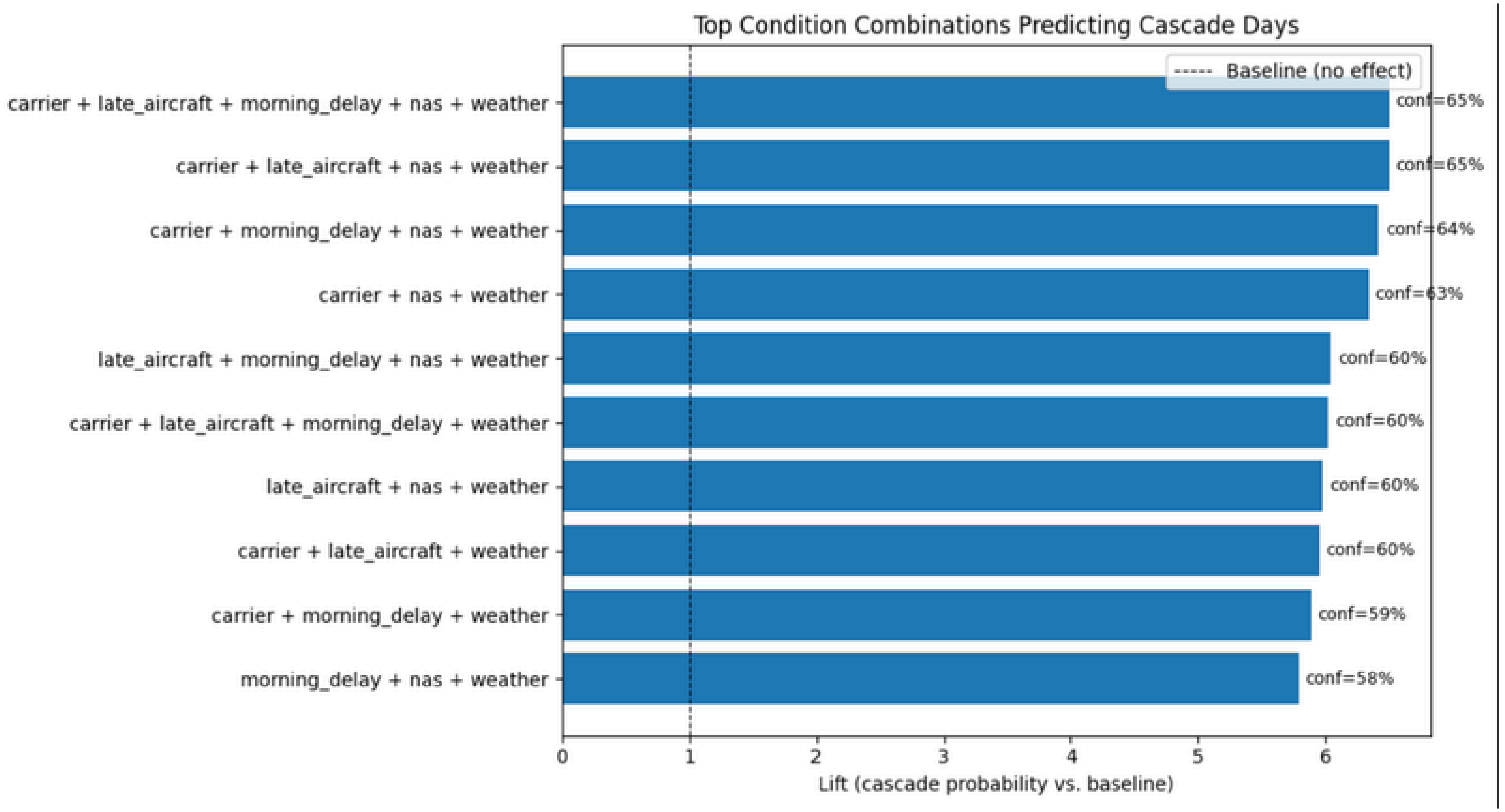
```
basket_df["high_weather"] = day_features["pct_weather_delay"] > day_features["pct_weather_delay"].quantile(0.75)
basket_df["high_late_aircraft"] = day_features["pct_late_aircraft"] > day_features["pct_late_aircraft"].quantile(0.75)
```

```
threshold = day_features["total_delay"].quantile(0.90)
day_features["is_cascade"] = (day_features["total_delay"] >= threshold).astype(int)
```

What combination of conditions reliably predicts large-scale cascades?

Cascades are not simply isolated delay causes. They are most predictable when multiple delays are stressed at the same time. Weather, though, does act as the most common trigger.

Apriori, found **zero** cascade predicting rules without weather.



Conclusion

- Delays build up and compound throughout the day, not independently
- Late aircraft delay dominates, thus showing that delays propagate from flight to flight
- Cascade failures are driven by multiple failures; there is no single cause
- Weather acts as a common trigger for cascading events
- A small set of conditions can predict cascades with high confidence.
- Apriori, found no combinations that could reliably predict cascade events unless weather was also present

Thank You